GROWTECH WALL

Connected power and control overview

GTW-E01 / REV J

07 NOV 2026 | AS-BUILT DOCUMENTATION DRAFT

LEFT: hydroponics NH1. RIGHT: gantry NG1. Component side, USB-C up: D13 top left / TX1 bottom right on both boards.

ROUTING GUIDE

Matching supply labels join.
Full hydro logic feeds: E03.
Full DC power routes: E04.
Only filled dots join crossings.

Both Nano headers: USB-C up.

D13 top left; TX1 bottom right.

CAM1
Logitech USB webcam
C920 per report

USB camera

DISP1
Pi touchscreen
Ribbon; separate supply untraced

ribbon

USB -> HYDROPONICS NH1

PI1
Raspberry Pi 5
Supervisor / vision / 9600-baud serial

USB host
Ribbon
AI HAT+ present; application use unconfirmed

27 / GPIO0 TX — orange
28 / GPIO1 RX — purple
25 / GND — grey

PI_USB_PWR

/dev/ttyAMA1 | 9600 baud

USB -> GANTRY NG1

ARM1
Hiwonder bus servo controller
Owner-identified model / reference E08

GND
TX
RX
5V / no Pi connection
+12V_LOAD — +12 V upstream
0V12_LOAD — 0V12

chain
base lead

SERVOS
5 reported servos
Bus ports: - / + / S (E08)
1 x HX-12H + 3 x HTS-35H
Reported 3-wire daisy chain
1 x HX-35M base
Separate 3-black-wire lead

+12V_LOAD
0V12_LOAD

RSPRAY / PUMP-S
Spray relay + 12 V pump
IN
Logic supply: trace

Spray: NH1 D10 / yellow

+24V_ACT
0V24_ACT

NH1
Hydroponics Nano
Component side / USB-C at top

D13, 3V3, AREF, A0, A1, A2, A3, A4, A5, A6, A7, 5V, RST, GND*, VIN
D12, D11, D10, D9, D8, D7, D6, D5, D4, D3, D2, GND, RST, RX0, TX1

X1 MOLEX
WIRE-ENTRY SIDE

Black end above / purple end below

0V12_LOAD	0V12	12 V load return
+12V_LOAD	+12V	+12 V load supply
NH1.GND	LOG GND	Sensor + relay logic GND
NH1.5V	+5V LOG	Sensor + relay logic +5 V
	EMPTY	
	LEVEL	Water-level interface OUT
	EC	EC interface OUT
	pH	pH interface OUT
	WATER	Water-pump relay IN
	FERT A	Fertilizer A relay IN
	FERT B	Fertilizer B relay IN
	ACID	Acid relay IN

DOSING MODULE
Sensors + four pump-relay functions
Colour correction: white D5 / light orange D4 (unverified).

12 V -> relay/PWM functions -> 4 pumps; exact contacts: E03

NG1
Gantry Nano
Component side / USB-C at top

D13, 3V3, AREF, A0, A1, A2, A3, A4, A5, A6, A7, 5V, RST, GND, VIN
D12, D11, D10, D9, D8, D7, D6, D5, D4, D3, D2, GND*, RST, RX0, TX1

DX / DM542
PUL+ — X / TOP — +24V
DIR+ — 0V24
ENA+
PUL-/DIR-/ENA-

MX / 4 wires / coil map untraced

DY1 / DM542
PUL+ — Y1 / LEFT — +24V
DIR+ — 0V24
ENA+
PUL-/DIR-/ENA-

MY1 / 4 wires / coil map untraced

DY2 / DM542
PUL+ — Y2 / RIGHT — +24V
DIR+ — 0V24
ENA+
PUL-/DIR-/ENA-

MY2 / 4 wires / coil map untraced

LX / X left end
LY1 / left bottom
LY2 / right bottom
Limit contacts / return untraced

A0 / A1 / A2: digital enable outputs
* GND net shared; occupied pad untraced.

* Shared GND net; installed pad untraced.

RTC1 | DS3231
SDA
SCL
VCC
GND

FAN / PD1 5 V output
+5V_FAN
0V_FAN

RLIGHT
LED relay box (photo E07)
IN: D9 (white)
NH1.5V — Logic +5 V from NH1
NH1.GND — Logic GND: colour / route ?
Load terminals: trace

LED1 / Grow-light group
5V lighting load
Load return

+12V_LOAD
0V12_LOAD

POWER

PSU1
24 V DC / 16.5 A
+24
0V24
AC L/N/PE + fuse: trace

ES1
Actuator interruption
IN +24 — OUT +24
IN RET — OUT RET

+24V_ACT
0V24_ACT

DC1
SZBK07 set to 12 V
IN+
IN-

DIM1
PWM light dimmer
Lighting load supply
IN 5V
IN RET

DC2
SZBK07 set to 5 V
OUT +5V
OUT RET

PD1
Wheeltec P23C / Pi + fan supply
USB-C to PI1; separate fan output:
+5V_FAN
0V_FAN

PI_USB_PWR

orange · purple · yellow · light blue · white · light orange · grey · green · red · black · brown

THICK = DC POWER THIN = SIGNAL TEAL = USB/RIBBON CONVENTION DASHED = UNRESOLVED TERMINALS / REFERENCE

Solid routes follow the owner record. Filled dots join; bridged crossings do not. DC returns are not protective earth.

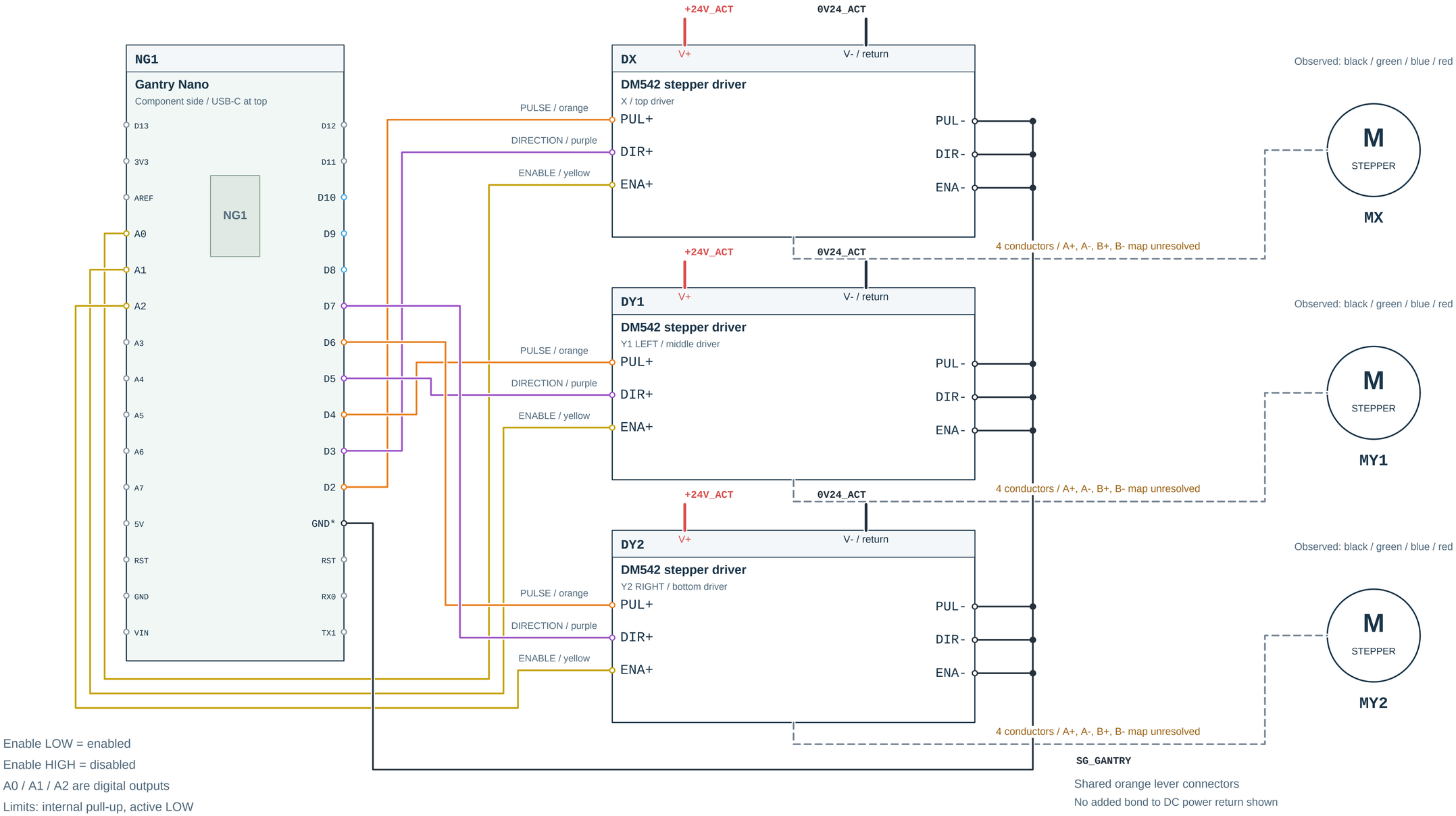
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Gantry motion, driver signals and limits

GTW-E02 / REV J

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NG1 component side, USB-C up: D13 top left and TX1 bottom right. Rails continue to E04; paired physical reference E05.



Enable LOW = enabled

Enable HIGH = disabled

A0 / A1 / A2 are digital outputs

Limits: internal pull-up, active LOW

LIMIT INPUTS / UNTRACED CONTACT BOUNDARY

NG1.D8

LX | X left end

Signal recorded; COM/NO/NC + return untraced

Expected assertion is LOW; contact type not assigned

NG1.D9

LY1 | Y1 left bottom

Signal recorded; COM/NO/NC + return untraced

Expected assertion is LOW; contact type not assigned

NG1.D10

LY2 | Y2 right bottom

Signal recorded; COM/NO/NC + return untraced

Expected assertion is LOW; contact type not assigned

orange · purple · yellow · light blue · white · light orange · grey · green · red · black · brown

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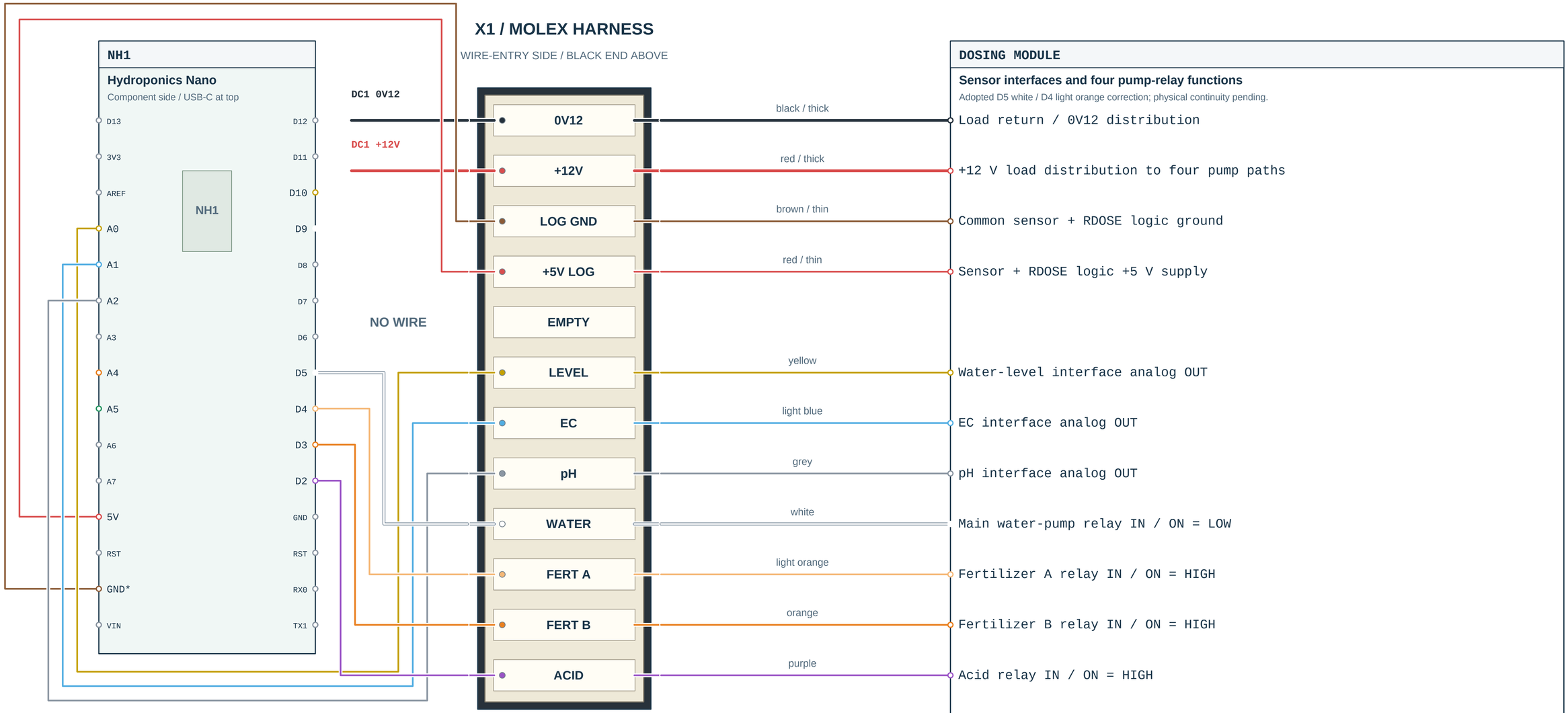
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Molex wire-entry layout and hydroponics connections

GTW-E03 / REV J

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NH1: component side, USB-C up, real two-header pin order. Molex: photographed wire-entry order. Crossovers without dots are not joins.



Both Nanos receive power over USB. A0 / A1 / A2 are analog inputs.

* GND pads share a net; the occupied pad is untraced (E05).

DIRECT FROM NH1: RTC1 on A4/SDA orange + A5/SCL green, with 5V red / GND black.

RLIGHT: D9 white, VCC from NH1 5V, logic-GND colour unresolved. RSPRAY: D10 yellow; logic supply / reference untraced.

+12V / 0V12 from X1

Fertilizer A / 12 V

Kamoer peristaltic

Relay/load terminals to trace

M

+12V / 0V12 from X1

Water pump / 12 V

Rotary PWM + relay route

Relay/load terminals to trace

M

+12V / 0V12 from X1

Fertilizer B / 12 V

Kamoer peristaltic

Relay/load terminals to trace

M

+12V / 0V12 from X1

Acid pump / 12 V

Kamoer peristaltic

Relay/load terminals to trace

M

orange · purple · yellow · light blue · white · light orange · grey · green · red · black · brown

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Solid routes follow the owner record. Filled dots join; bridged crossings do not. DC returns are not protective earth.

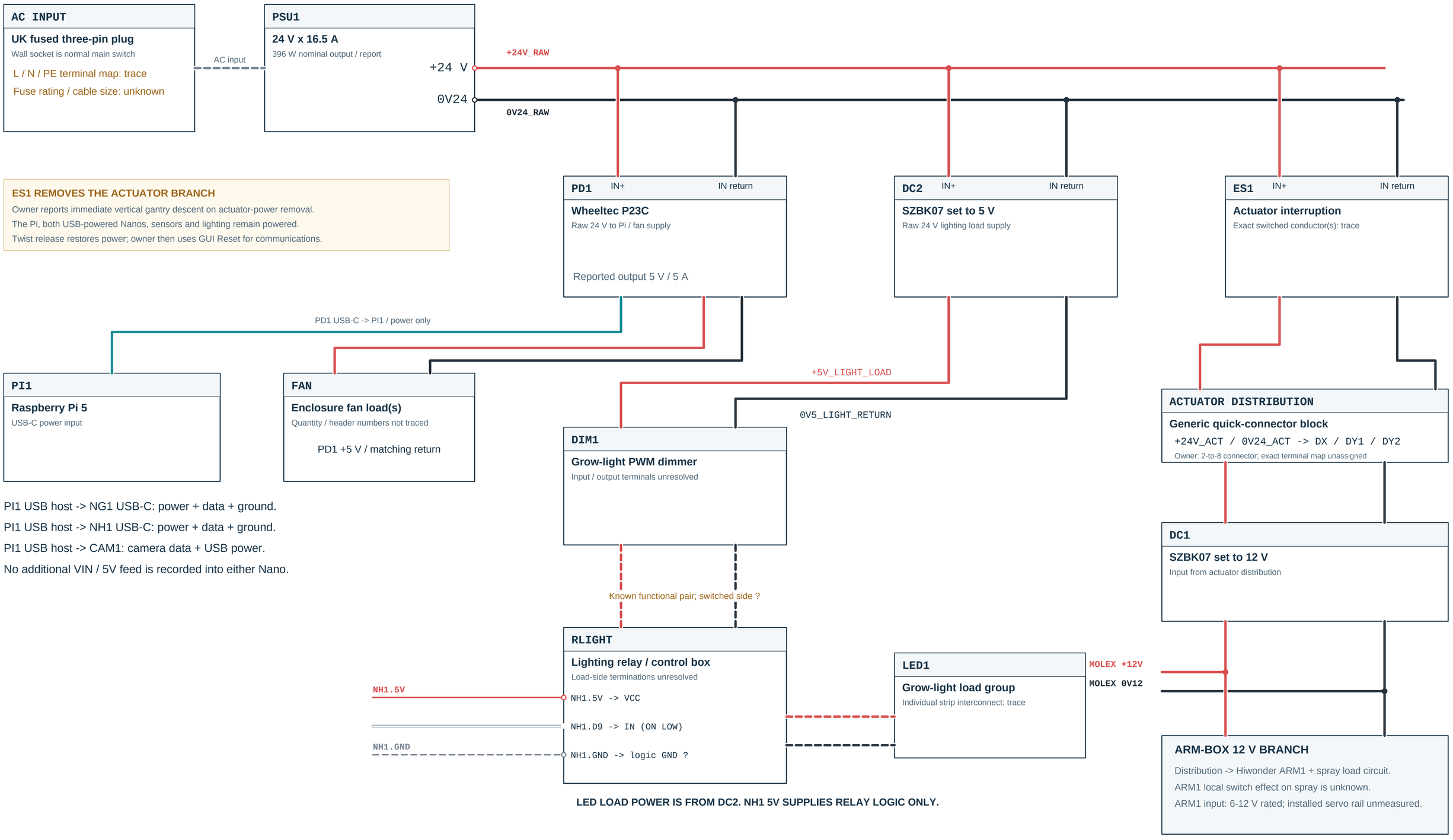
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Power distribution, lighting and arm-box branches

GTW-E04 / REV J

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Supply and corresponding returns are paired. No unverified DC ground bond, protective-earth bond, fuse value or relay contact assignment is added.



orange purple yellow light blue white light orange grey green red black brown

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Solid routes follow the owner record. Filled dots join; bridged crossings do not. DC returns are not protective earth.

GROWTECH WALL

Physical header positions and connector identification

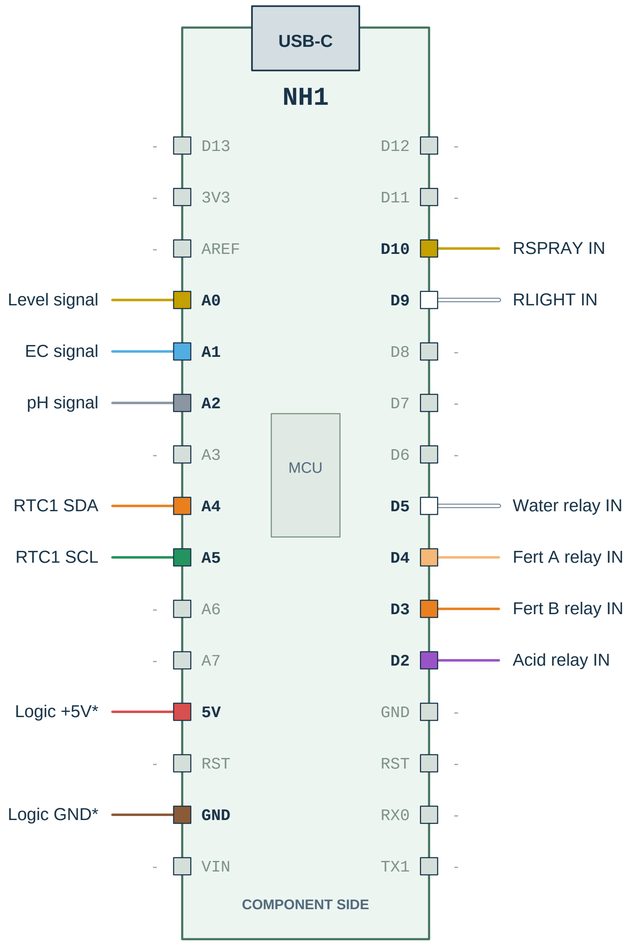
GTW-E05 / REV J

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Nano component side, USB-C at top. Nano footprint is a reference matching the owner count; verify the actual clone silkscreen before rewiring.

BOTH NANOS: USB-C UP / D13 TOP LEFT / TX1 BOTTOM RIGHT

NH1 / HYDROPONICS



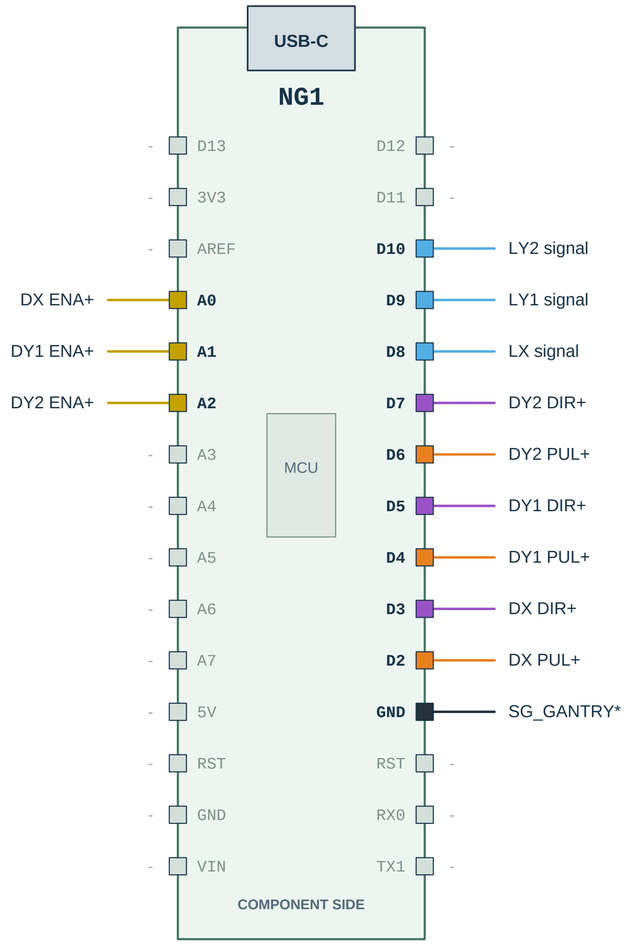
* NH1 5V -> X1.+5V, RTC1 VCC and RLIGHT VCC.

X1.GND is brown; RTC1 return is black; RLIGHT return colour is unknown.

Both on-board GND pads are equivalent; the specific installed pad is untraced.

D5 white / D4 light orange: adopted colour correction, unverified.

NG1 / GANTRY

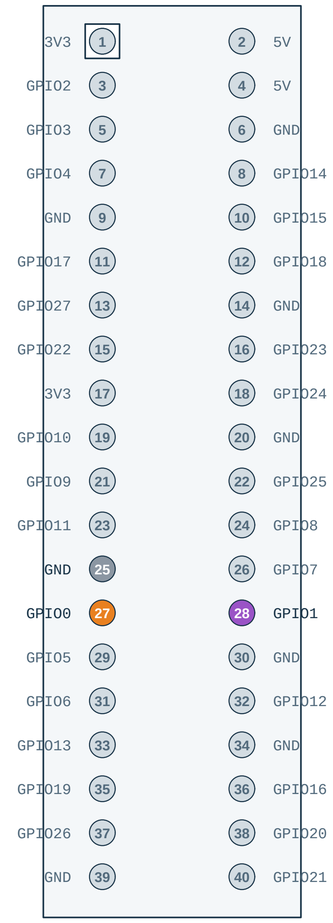


* GND pins share the same board net. The occupied header pad is untraced.

The drawing pin names are electrical identifiers; unlabelled pins have no recorded wire.

The USB-C-equipped clones are not identified as official Arduino A000005 hardware.

PI1 / 40-PIN HEADER



Mating header view / pin 1 at upper left

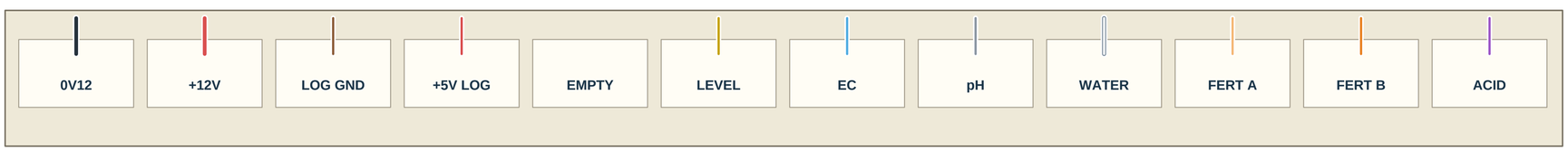
```
25  GND       -> ARM1 GND   grey
27  GPIO0 TX  -> ARM1 RX    orange
28  GPIO1 RX  <- ARM1 TX    purple
```

UART1 alternate function supported by RP1.

Owner port: /dev/ttyAMA1 at 9600 baud.

Pi logic is 3.3 V; ARM1 TX voltage unverified.

X1 / MOLEX WIRE-ENTRY SEQUENCE (BLACK END LEFT, PURPLE END RIGHT)



Same single row as E03, rotated for readability. Photo E07 anchors the wire-entry view; use function labels on this side.

orange | purple | yellow | light blue | white | light orange | grey | green | red | black | brown

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GROWTECH WALL

Connection schedule, evidence limits and verification register

GTW-E06 / REV J

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Use this sheet with E01-E05. The separate CSV contains the complete recorded connection schedule and endpoint names.

SIGNAL SCHEDULE / DIRECT LINKS

SOURCE	DESTINATION	WIRE / FUNCTION
PI1 USB host	NG1 USB-C	Power + data + GND / gantry
PI1 USB host	NH1 USB-C	Power + data + GND / hydroponics
PI1 USB host	CAM1 USB	Camera data + USB power
PI1 ribbon	DISP1	Ribbon link / power route untraced
PI1 pin 27 / GPIO0 TX	ARM1 RX	Orange / UART1
PI1 pin 28 / GPIO1 RX	ARM1 TX	Purple / UART1
PI1 pin 25	ARM1 GND	Grey / UART reference
NH1 A4 / SDA	RTC1 SDA	Orange / I2C
NH1 A5 / SCL	RTC1 SCL	Green / I2C
NH1 5V / GND	RTC1 VCC / GND	Red / black
NH1 5V	RLIGHT VCC	Red supply callout / latest correction
NH1 D9	RLIGHT IN	White / firmware ON LOW
NH1 GND	RLIGHT logic GND	Return colour + termination unresolved
NH1 D10	RSPRAY IN	Yellow / firmware ON HIGH

HOW TO READ THIS SET

A filled junction dot joins conductors; crossings with white bridges are not joins.
Dashed boundaries retain known functional routes without inventing hidden terminals.
Every repeated net name on a detail sheet identifies the same electrical net.
Power-return and signal-ground names remain distinct unless the supplied evidence joins them.
No fresh hardware continuity tests or live firmware checks were performed.

MANUFACTURER REFERENCE CHECKS

[1] Arduino Nano full pinout, A000005, p.1: standard 30-pin signal-name order.
https://docs.arduino.cc/resources/pinouts/A000005-full-pinout.pdf
[2] Raspberry Pi RP1 Peripherals, Table 4, printed p.15: GPIO0/1 UART1 TX/RX.
https://datasheets.raspberrypi.com/rp1/rp1-peripherals.pdf
[3] Raspberry Pi hardware / configuration documentation: header numbering; 3.3 V UART.
https://www.raspberrypi.com/documentation/computers/raspberry-pi.html

TERMINAL VERIFICATION REGISTER

V01 AC INPUT / EARTH
PSU model; L/N/PE; plug fuse; cable size; protective-earth path.

V02 ES1 POWER INTERRUPTION
Identify switched conductor(s), poles and actual contact terminals.

V03 DM542 / MOTORS
Match installed variant, coil pairs, A+/A-/B+/B- and DIP orientation.

V04 LIMIT SWITCHES
Identify actual signal contact, COM/NO/NC selection and return route.

V05 X1 CONNECTOR
D5 white / D4 light orange is an adopted hypothesis; confirm continuity and mating orientation.

V06 DOSING CONTACTS / PWM
Trace relay load contacts and main-pump PWM terminal sequence.

V07 LIGHTING LOAD / LOGIC
Trace DIM1 / RLIGHT contacts, return switching and logic-GND wire.

V08 ARM UART / SERVOS
Hiwonder controller identified; verify TX level, servo rail, occupied sockets, IDs and chain order.

V09 SPRAY SWITCHING
Identify logic VCC / GND and pump contact path; switch interaction.

V10 SENSOR / PERIPHERAL DETAILS
Sensor interface models; fan terminals / count; touchscreen power.

DRAWING BASIS

Primary evidence: attached GrowTech-Wall-Schematic-Drawing-Brief.md (03 Nov 2026).
New owner photos: LED relay box, single-row Molex connector and robotic arm control box.
Rev J: ARM1 identified as Hiwonder Serial Bus Servo Controller; physical reference added on E08.
External box identities confirmed by owner; obscured internal terminals remain unresolved.

References establish pin capabilities, not hidden wiring on this prototype.
Hiwonder controller references and photo pin order: E08.

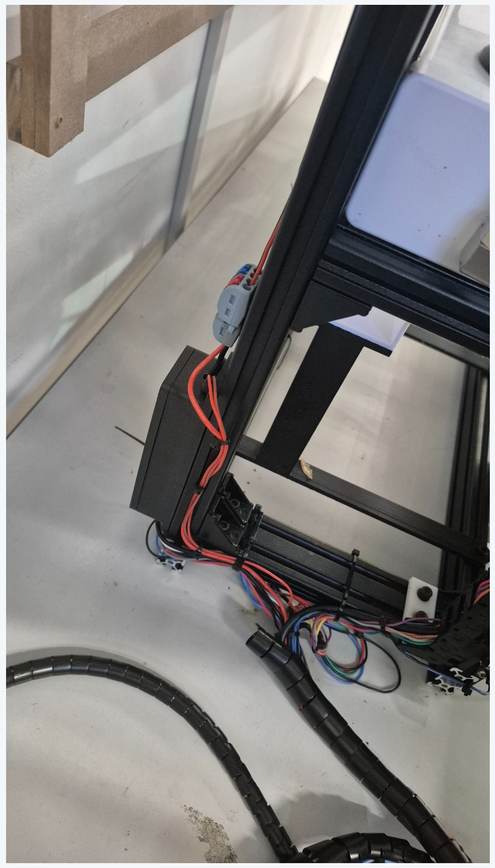
orange purple yellow light blue white light orange grey green red black brown

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Owner identifications: first photo = LED relay box; second = Molex connector; third = robotic arm control box. Photos are reference evidence.

LED RELAY BOX

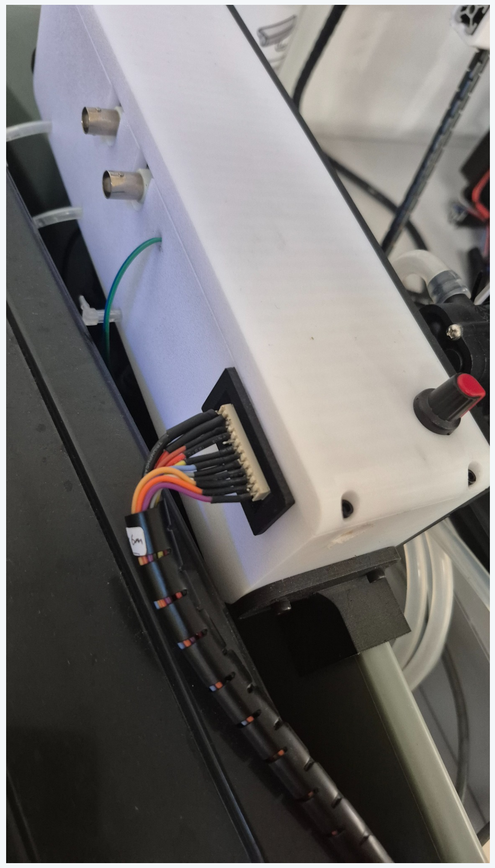


Black enclosure mounted behind the frame.

Load leads and control harness enter the box.

Internal contacts / logic return remain obscured.

MOLEX CONNECTOR

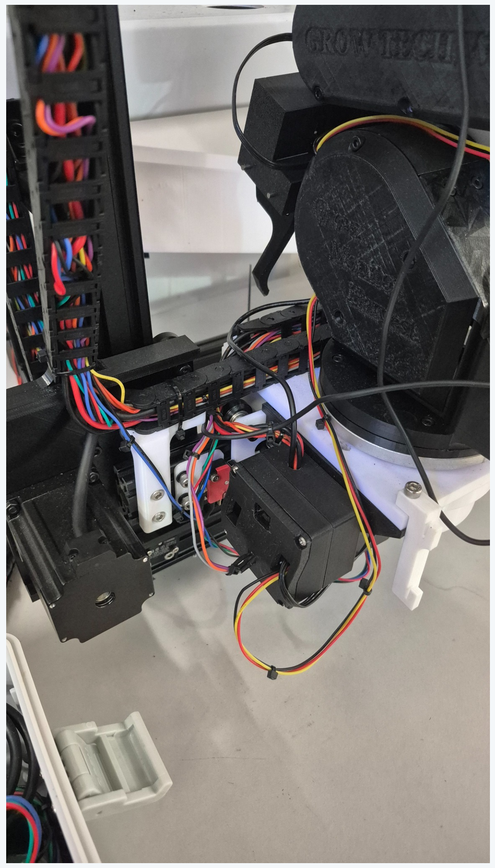


Single row of wire entries on the dosing box.

Black return end -> purple acid-signal end.

Heat-shrink covers parts of individual wires.

ROBOTIC ARM CONTROL BOX



Black enclosure beside the arm base.

External red local switch and lead groups visible.

ARM1 model identified separately; board view E08.

MOLEX: SINGLE ROW, WIRE-ENTRY SIDE, ROTATED HORIZONTALLY FOR READING

Wire-entry view: black return end to purple acid end. White D5 / light orange D4 mapping is adopted provisionally.

0V12	+12V	LOG GND	+5V LOG	EMPTY	LEVEL	EC	pH	WATER	FERT A	FERT B	ACID
black	red	brown	red	no wire	yellow	light blue	grey	white	light orange	orange	purple

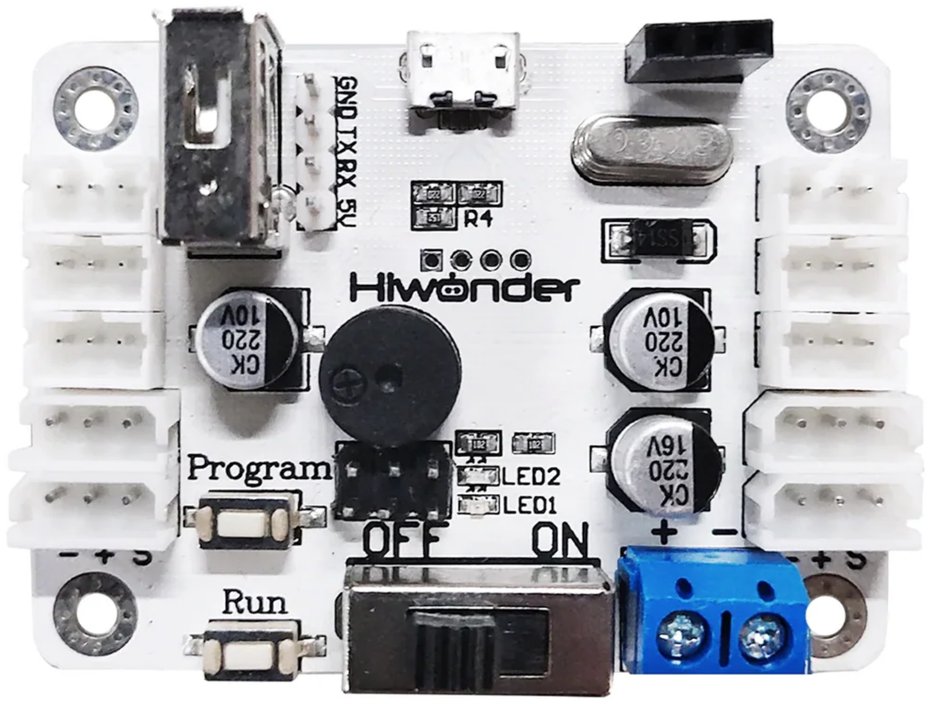
orange · purple · yellow · light blue · white · light orange · grey · green · red · black · brown

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Owner-identified Hiwonder Serial Bus Servo Controller. Component-side reference photo; micro-USB at top and power switch at bottom.

SUPPLIED BOARD REFERENCE / ARM1



The product photo identifies the board; it does not show the installed harness.

SERVO CONNECTIONS / WHAT IS NOW IDENTIFIED

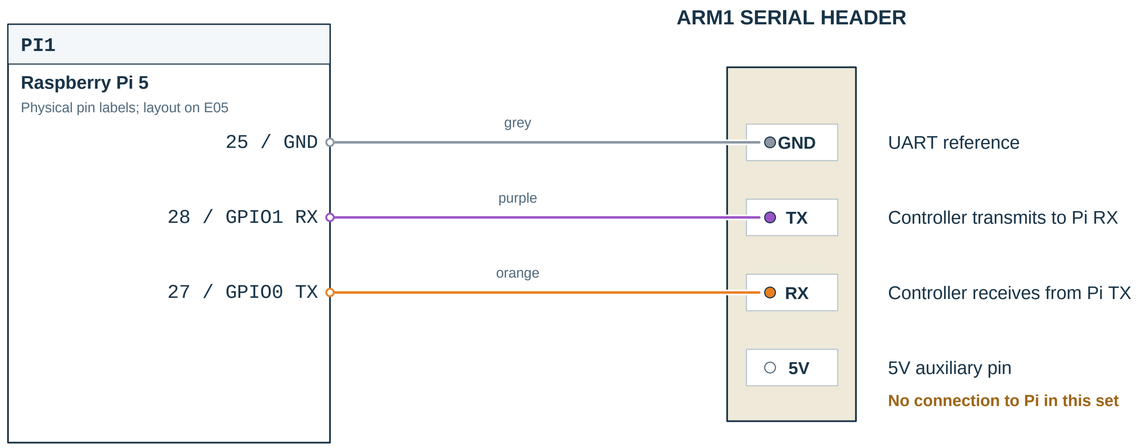
Visible three-pin port marking: - / + / S (signal), left to right in this component-side photo.

The recorded four-servo chain and separate base lead remain unchanged.

Installed socket choice, servo-ID order and cable-contact mapping still require tracing.

UART CONNECTION / 9600 BAUD

Serial header order in the supplied photo, from top to bottom:
GND / TX / RX / 5V. TX and RX are named from the controller perspective.

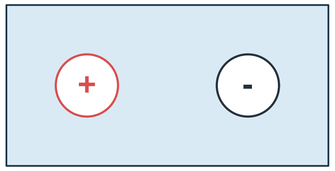


The host UART is distinct from the three-wire servo bus.

Keep the recorded Pi GPIO0 / GPIO1 assignment; no GPIO remapping is implied.

Hiwonder documents a Raspberry Pi serial connection. Installed TX voltage has not been measured.

MAIN POWER TERMINAL / PHOTO ORIENTATION



DC1 +12V_LOAD

DC1 0V12_LOAD

Left screw = +; right screw = - in this view.

Manufacturer input range: 6-12 V.

Recorded DC1 branch: 12 V.

Select supply voltage to suit the connected servos.

This is separate from the serial-header 5V pin.

REFERENCES / OWNER IMAGE + HIWONDER DOCUMENTATION

[4] Product: Serial Bus Servo Controller, SKU 21090003.
https://www.hiwonder.com/products/serial-bus-servo-controller

[5] Bus Servo Controller User Manual, sections 1.2.1-1.2.2.

[6] Bus Servo Controller Secondary Development, sections 2.1 and 2.5.

Official manual and development URLs are included in the editable package.

Reference pin capabilities do not establish hidden wiring inside the arm box.